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ODD END SEMESTER EXAMINATION DECEMBER-2025

Name of the Course: B.Tech

Name of the Course: Engineering Chemistry

Time: 3 Hours

Semester: I

Course Code: TCH-101

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	What is the molecular orbital theory? With the help of MO diagram calculate the bond order and magnetic behavior of the CO molecule	CO1
(b)	Define the basic principle of the UV spectroscopy. Describe the various electronic transitions possible in UV Spectroscopy	
(c)	What is meant by intermolecular and intramolecular H-bonding? Explain why (i) H ₂ O is liquid but H ₂ S is gas (ii) acetone is more volatile than alcohol	
Q2	(20 marks)	
(a)	State the zeolite process for the removal of the hardness of water. Discuss it's merit over lime soda process	CO2
(b)	Elaborate the L-S process and Reverse osmosis process of the water softening.	
(c)	The hardness of 5000 liter of water sample was removed by passing it through a zeolite softener, the zeolite softener required 50 liter of NaCl solution containing 2.5 g/L of NaCl for regeneration. Find out the hardness of water sample.	
Q3	(20 marks)	
(a)	Explain how polymers are classified based on their structure and mode of polymerization	CO3
(b)	Explain the term conducting polymer in terms characteristics, classification and applications.	
(c)	Evaluate the free radical mechanism of addition polymerization reaction	
Q4	(20 marks)	
(a)	Describe the characteristics of a good fuel and explain the terms gross calorific value and net calorific value.	CO4
(b)	A 0.90 gm sample of solid fuel was completely combusted in the excess of oxygen using bomb calorimeter. the rise in temperature of water in calorimeter was 2.5 °C calculate the high calorific value of the fuel if water taken in calorimeter is 3000 g and water equivalent of calorimeter is 3200 g also calculate low calorific value (% of hydrogen in fuel=2.2)	
(c)	How the calorific value of the fuel is is determined by using the bomb calorimeter.	
Q5	(20 marks)	
(a)	What is galvanic cell? Discuss it's working with the suitable diagram	CO5
(b)	What is an electrochemical series? Discuss it's important applications	
(c)	Define corrosion. Explain the rusting of iron with the help of electrochemical theory of corrosion.	

